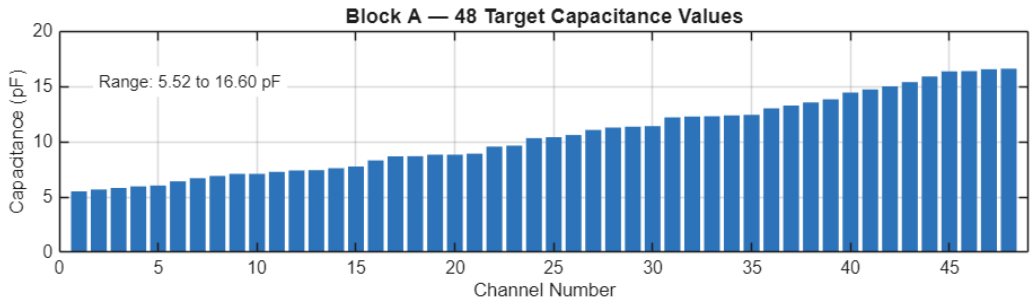
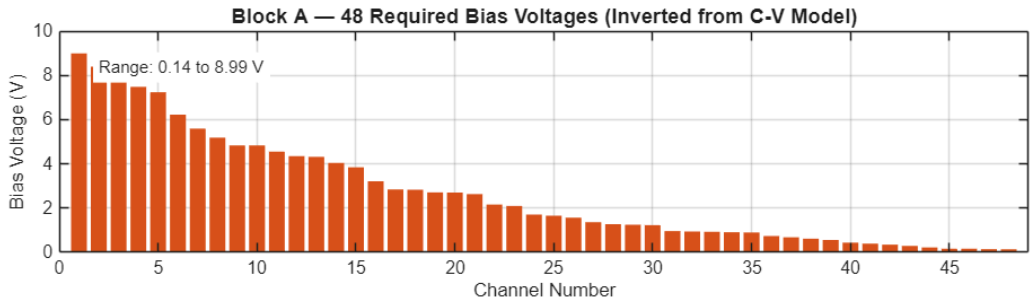
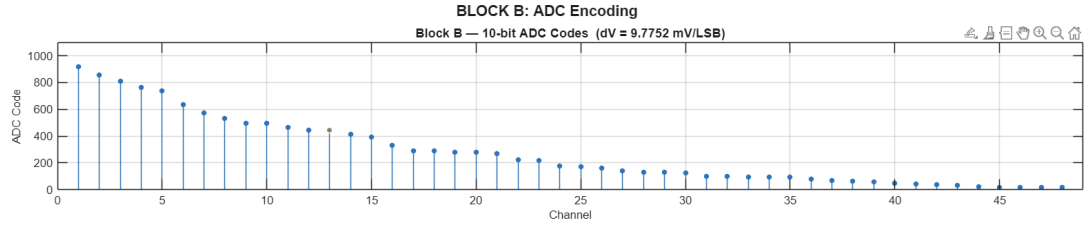
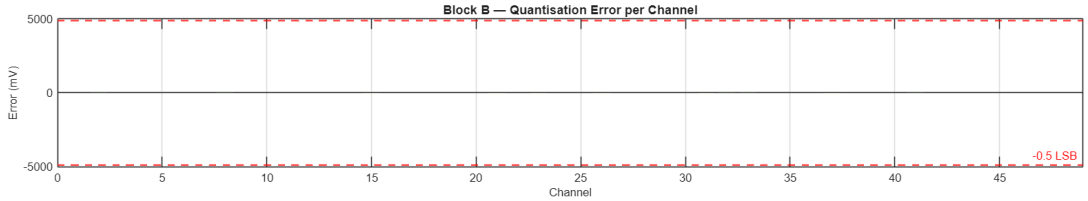
BLOCK A: Transmitter Input Stage

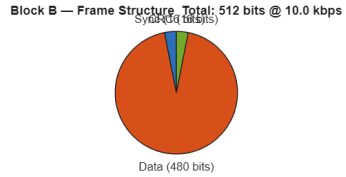




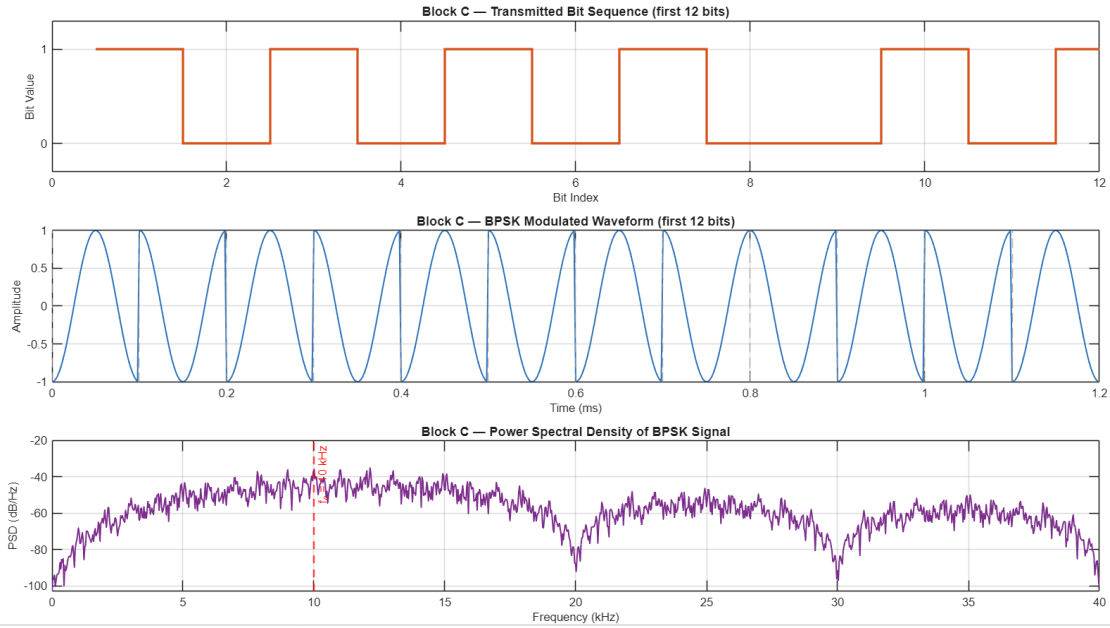
BLOCK B: ADC Encoding



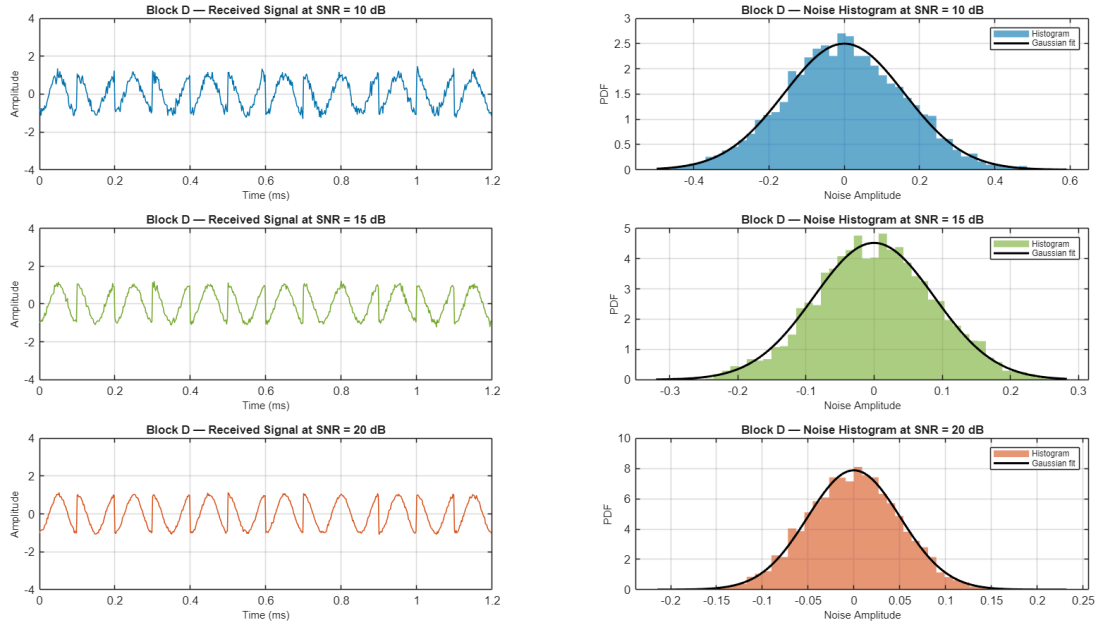




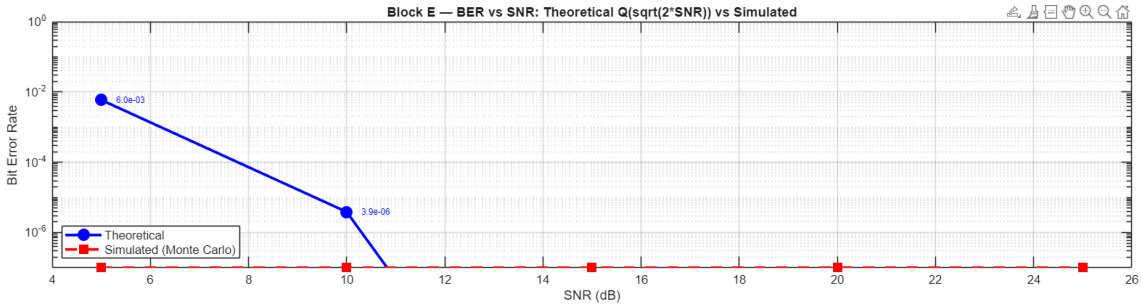
BLOCK C: BPSK Digital Modulation

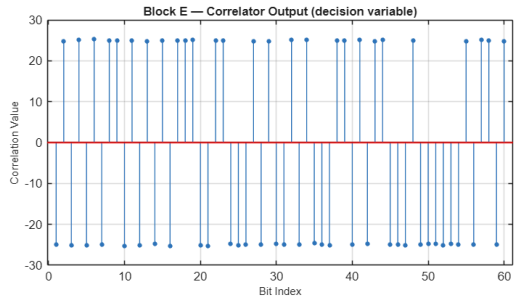


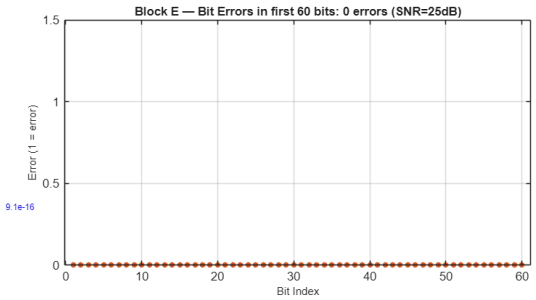
BLOCK D: AWGN Channel Model



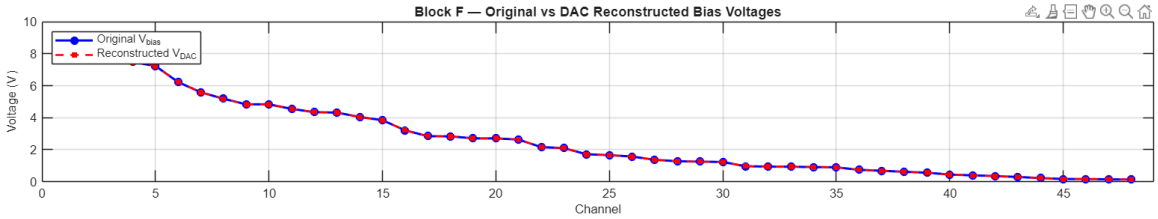
BLOCK E: Demodulation & Bit Recovery

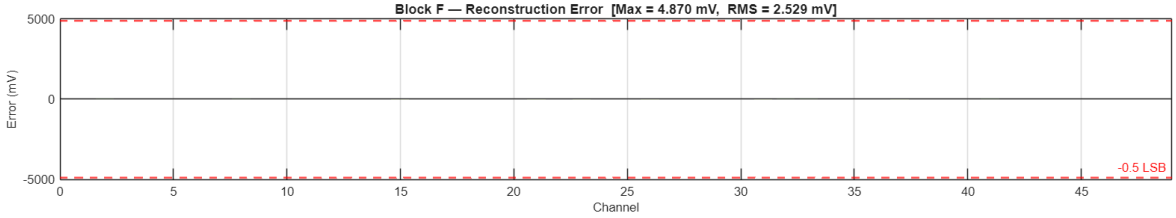


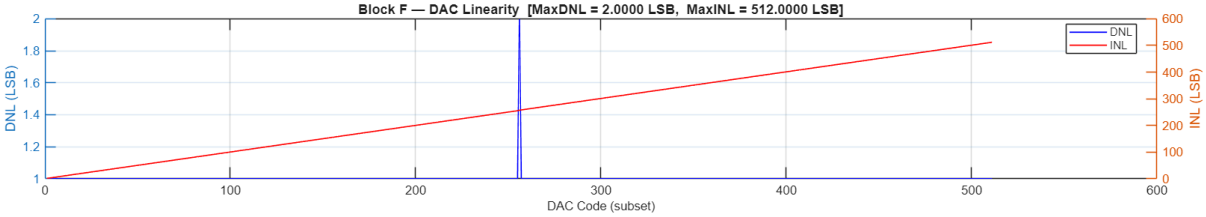




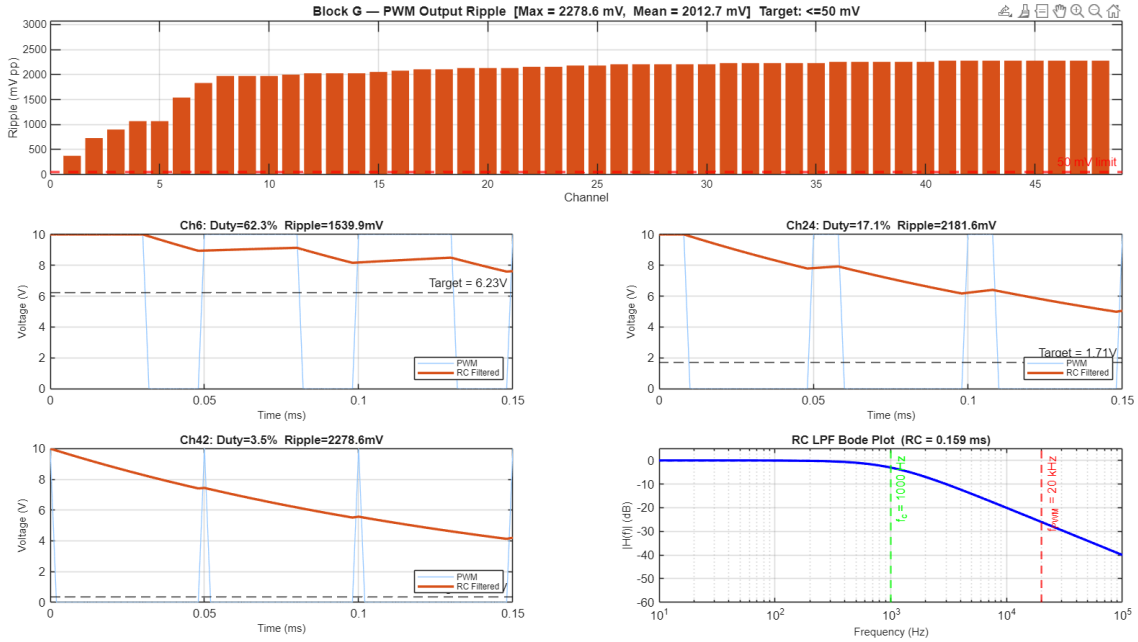
BLOCK F: DAC Reconstruction



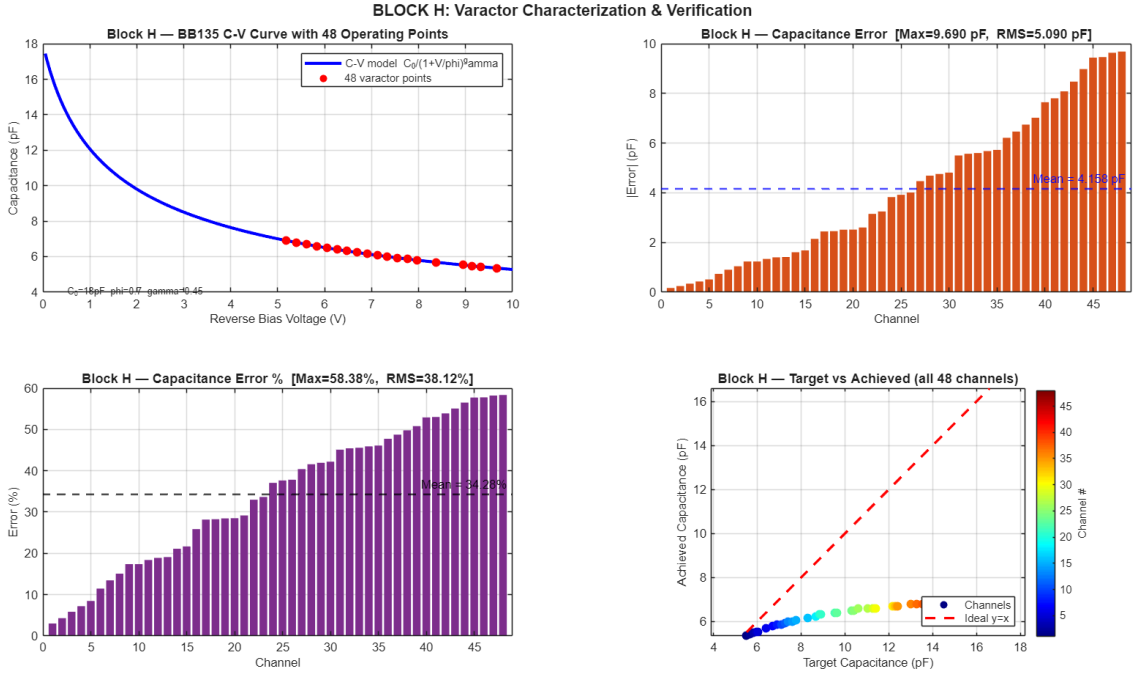




BLOCK G: PWM-Based DC Bias Generation



BLOCK H: Varactor Characterization & Verification



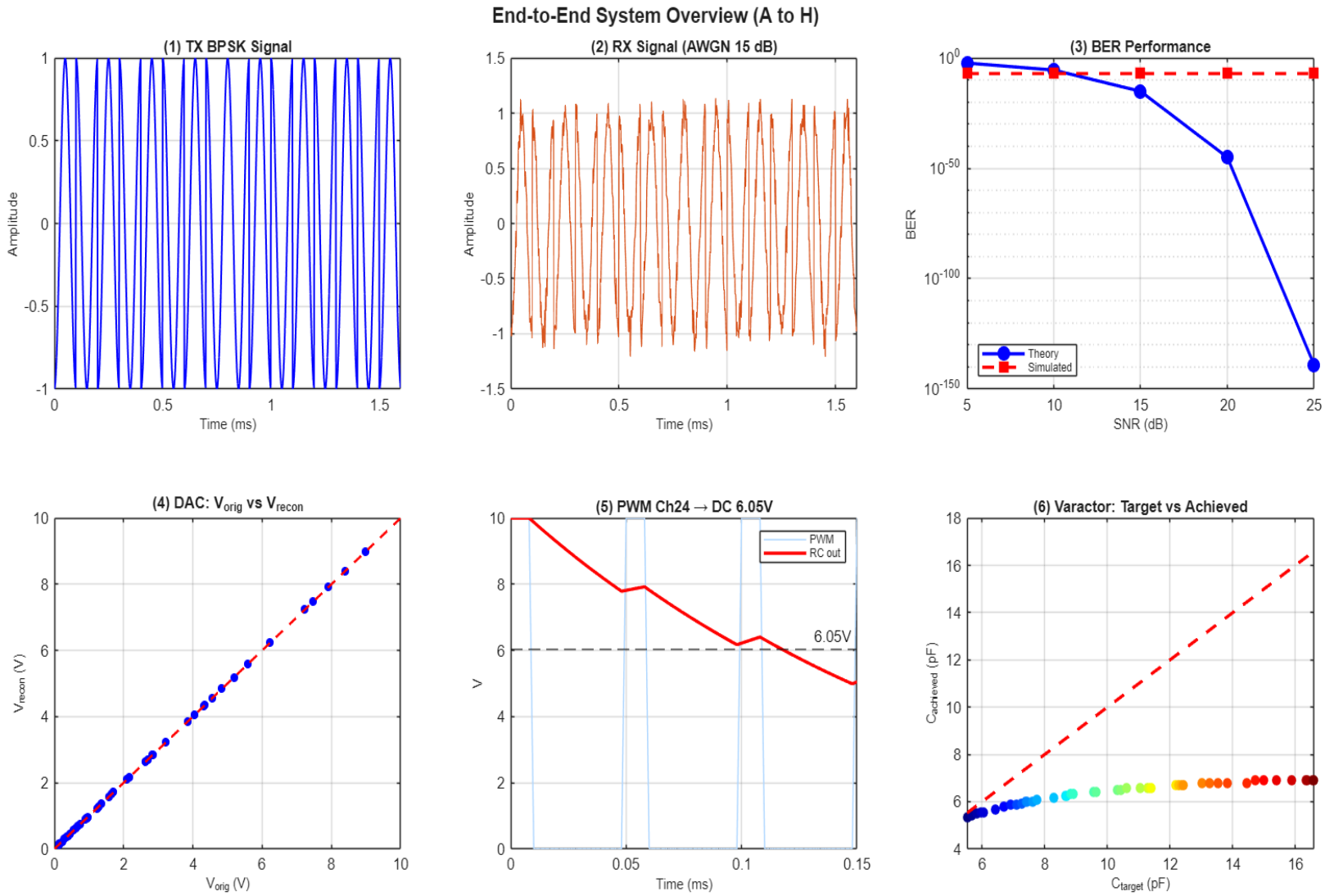

End-to-End System Overview (A to H)
(1) TX BPSK Signal
Amplitude
Time (ms)
(2) RX Signal (AWGN 15 dB)
Amplitude
Time (ms)
(3) BER Performance
BER
SNR (dB)
Theory
Simulated
(4) DAC: V_{orig} vs V_{recon}
V_{recon} (V)
V_{orig} (V)
(5) PWM Ch24 → DC 6.05V
PWM
RC out
6.05V
V
Time (ms)
(6) Varactor: Target vs Achieved
$C_{achieved}$ (pF)
C_{target} (pF)

FIGURE 10 — Full 48-Channel End-to-End Results Table

Ch	C target (pF)	V bias (V)	ADC Co...	V recon (V)	Ripple (mV...	C achieved (pF)	Error (pF)	Error (%)
1	5.5168	8.9918	920	8.9932	374.2100	5.3518	0.1650	2.9900
2	5.6779	8.3911	858	8.3871	729.9500	5.4349	0.2430	4.2800
3	5.8187	7.9094	809	7.9081	900.9600	5.4796	0.3391	5.8280
4	5.9545	7.4792	765	7.4780	1.0674e+03	5.5266	0.4279	7.1870
5	6.0359	7.2362	740	7.2336	1.0674e+03	5.5266	0.5093	8.4370
6	6.4167	6.2274	637	6.2268	1.5399e+03	5.6826	0.7341	11.4410
7	6.7011	5.5908	572	5.5914	1.8326e+03	5.8007	0.9004	13.4370
8	6.9049	5.1857	530	5.1808	1.9724e+03	5.8646	1.0403	15.0660
9	7.0975	4.8366	495	4.8387	1.9724e+03	5.8646	1.2329	17.3710
10	7.0978	4.8361	495	4.8387	1.9724e+03	5.8646	1.2332	17.3740
11	7.2671	4.5535	466	4.5552	1.9997e+03	5.9321	1.3351	18.3710
12	7.3991	4.3476	445	4.3500	2.0267e+03	6.0023	1.3968	18.8780
13	7.4175	4.3198	442	4.3206	2.0267e+03	6.0023	1.4152	19.0790
14	7.6074	4.0456	414	4.0469	2.0267e+03	6.0023	1.6051	21.1000
15	7.7553	3.8468	394	3.8514	2.0533e+03	6.0754	1.6798	21.6610
16	8.2974	3.2129	329	3.2160	2.0796e+03	6.1517	2.1457	25.8600
17	8.6762	2.8433	291	2.8446	2.1056e+03	6.2314	2.4448	28.1780
18	8.6869	2.8337	290	2.8348	2.1056e+03	6.2314	2.4555	28.2660
19	8.8281	2.7093	277	2.7077	2.1312e+03	6.3147	2.5134	28.4700
20	8.8325	2.7056	277	2.7077	2.1312e+03	6.3147	2.5177	28.5050
21	8.9153	2.6356	270	2.6393	2.1312e+03	6.3147	2.6006	29.1700
22	9.5533	2.1607	221	2.1603	2.1566e+03	6.4019	3.1514	32.9870
23	9.6488	2.0982	215	2.1017	2.1566e+03	6.4019	3.2468	33.6500
24	10.3189	1.7103	175	1.7107	2.1816e+03	6.4934	3.8255	37.0730
25	10.4147	1.6613	170	1.6618	2.1816e+03	6.4934	3.9213	37.6520
26	10.6005	1.5703	161	1.5738	2.2063e+03	6.5894	4.0111	37.8390
27	11.0571	1.3673	140	1.3685	2.2063e+03	6.5894	4.4676	40.4050
28	11.2795	1.2777	131	1.2805	2.2063e+03	6.5894	4.6901	41.5810
29	11.3476	1.2515	128	1.2512	2.2063e+03	6.5894	4.7582	41.9310
30	11.4024	1.2307	126	1.2317	2.2063e+03	6.5894	4.8130	42.2100
31	12.1922	0.9637	99	0.9677	2.2307e+03	6.6904	5.5018	45.1260
32	12.2651	0.9418	96	0.9384	2.2307e+03	6.6904	5.5747	45.4520
33	12.2938	0.9333	95	0.9286	2.2307e+03	6.6904	5.6034	45.5790
34	12.3688	0.9114	93	0.9091	2.2307e+03	6.6904	5.6784	45.9090
35	12.4191	0.8969	92	0.8993	2.2307e+03	6.6904	5.7287	46.1280
36	13.0106	0.7400	76	0.7429	2.2548e+03	6.7968	6.2138	47.7600
37	13.2641	0.6796	70	0.6843	2.2548e+03	6.7968	6.4673	48.7580
38	13.5424	0.6174	63	0.6158	2.2548e+03	6.7968	6.7456	49.8110
39	13.8216	0.5590	57	0.5572	2.2548e+03	6.7968	7.0248	50.8250
40	14.4424	0.4419	45	0.4399	2.2548e+03	6.7968	7.6457	52.9390
41	14.7135	0.3956	40	0.3910	2.2786e+03	6.9091	7.8044	53.0430
42	14.9942	0.3506	36	0.3519	2.2786e+03	6.9091	8.0851	53.9220
43	15.3880	0.2918	30	0.2933	2.2786e+03	6.9091	8.4789	55.1010
44	15.8917	0.2233	23	0.2248	2.2786e+03	6.9091	8.9826	56.5240
45	16.3536	0.1663	17	0.1662	2.2786e+03	6.9091	9.4445	57.7520
46	16.3749	0.1638	17	0.1662	2.2786e+03	6.9091	9.4658	57.8070
47	16.5490	0.1437	15	0.1466	2.2786e+03	6.9091	9.6400	58.2510
48	16.5990	0.1381	14	0.1369	2.2786e+03	6.9091	9.6899	58.3760